Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Global Production Center
GLOBAL_ANALYSIS_FORECAST_WAV_001_027

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	January 2018	All	Creation of the document.	Lotfi Aouf	Marie Drévillon, Angélique Melet
2.0	January 2018		Adjustment of the physics of the wave model MFWAM.	Lotfi Aouf	
2.1	Dec 2020		Use of SWIM-nadir data in the assimilation and extension of forecast.	Lotfi Aouf	Marie Drévillon
3.0	Sep 2022		Update of the assimilation with SWIM wave spectra and Sentinel-6 SWH.	Lotfi Aouf	Marie Drévillon
3.1	Nov 2023		Addition of a new variable: maximum crest-trough wave height.	Alice Dalphinet	Marie Drévillon
3.2	Nov 2024	I.1	Addition of a new variable: maximum crest wave height.	Alice Dalphinet	Olivier Le Galloudec

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I.EXECUTIVE SUMMARY

I.1 Products covered by this document

This document is the quality information document for the Copernicus Marine Environment Monitoring Service (CMEMS) global analysis and forecast wave product **GLOBAL_ANALYSIS_FORECAST_WAV_001_027**. This product provides aggregated analyses updated daily and 10-day forecasts, all defined on a standard grid at 1/10 degree (approx. 10km) resolution. A 2-years rolling archive is stored on the CMEMS server. The assimilation system is updated with SWIM-nadir (Surface Waves Investigation and Monitoring instrument), Sentinel 6 and SWOT-nadir (Surface Water and Ocean Topography) Significant Wave Height (SWH) and wave spectra at beam 10°of CFOSAT mission (Chinese-French Oceanic SATellite).

The product is organised as 1 dataset (**cmems_mod_glo_wav_anfc_0.083deg_PT3H-i**) containing 3-hourly instantaneous values for the following variables:

- Spectral significant wave height (Hm0)
- Maximum crest-trough wave height (VCMX)
- Maximum crest wave height (VMXL)
- Spectral moments (-1,0) wave period (Tm-10)
- Spectral moments (0,2) wave period (Tm02)
- Wave period at spectral peak / peak period (Tp)
- Mean wave direction from (Mdir)
- Wave principal direction at spectral peak
- Stokes drift U
- Stokes drift V
- Spectral significant wind wave height
- Spectral moments (0,1) wind wave period
- Mean wind wave direction from
- Spectral significant primary swell wave height
- Spectral moments (0,1) primary swell wave period
- Mean primary swell wave direction from
- Spectral significant secondary swell wave height
- Spectral moments (0,1) secondary swell wave period
- Mean secondary swell wave direction from

The products are described in more details in the Product User Manual CMEMS-GLO-PUM-001-027[1] ().

The scientific assessment and quality control were performed on a subset of variables that can be compared to altimetry and in situ observations:

- Spectral significant wave height (Hm0) *-> comparable with altimetry and buoy observations;*
- Wave period at spectral peak / peak period (Tp) *-> comparable with buoy observations;*

[1] http://marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-027.pdf (accessed 29/08/2024)

I.2 Summary of the results

I.2.1 Summary of the results related to update with SWIM directional wave spectra and SWH from Sentinel-6A (Version 4.2)

The directional wave observations from the CFOSAT mission have shown their capacity to significantly improve the wave physics, particularly in unlimited fetch conditions such as the storm conditions in the Southern Ocean (Aouf et al., 2021). The assimilation system of CMEMS-GLO-WAVES has been updated with significant wave height (SWH) from the Sentinel-6A altimeter mission and also with directional wave spectra from the SWIM instrument of the CFOSAT mission. The increasing number of assimilated data in the system led to a significant improvement of the integrated wave parameters. The comparison with independent SWH from HY2B and HY2C altimeters during June 2022 indicated an average scatter index of 7.9% and a small bias of 3 cm. The scatter index of SWH for high latitudes, mid latitudes, and the tropics were 8%, 7.8%, and 7.4 %, respectively.

I.3 Estimated Accuracy Numbers after the update with SWIM nadir data

To assess the update of the assimilation system with SWIM-nadir data, a period for validation starting from January to May 2020 was considered. The **global scale estimated accuracy numbers** were based on comparisons with **in situ wave observations from buoys (Table 1)**.

Table 1: EANs for GLOBAL_ANALYSIS_FORECAST_WAV_001_027 .

Variable	Metrics	Units	Value
Significant wave height	Global average of bias	cm	2.0
	Global average of normalized scatter index	%	13.8
Peak wave period	Global average of bias	seconds	0.33
	Global average of normalized scatter index	%	12.6

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: Météo-France, France

The sub-sections below provide a brief summary of the changes that were made in each update to the production system. This resulted in the release of a new version of the system, the validation of which was documented in the related QUID. The current system version is 4.2 and its validation is documented in this QUID V3.2.

Description of the change in V4.2 :

The assimilation system of CMEMS-GLO-WAV is updated with directional wave spectra from CFOSAT and SWH from the altimeter mission S6A. After the update, CMEMS-GLO-WAV uses two sources of directional wave spectra from SAR of Sentinel-1A and SWIM of CFOSAT. Sentinel-1B does not provide wave spectra as it is no longer operational having reached the end of its mission in late 2022. The SWIM instrument is able to capture directional wave spectra of waves for wavelength ranges from 60 to 500 m. Comparisons to Sentinel-1 CFOSAT will provide more information on shorter scale of waves which are still dominated by wind.

Description of the change in V4.1:

The CMEMS-GLO-WAV has been updated with more recent computing code ECWAM-IFS-41R2 (European Center WAve Model – Integrated Forecasting System). This allows a faster computation skill and the use of GRIB_API libraries for read/write routines. There is no change in the wave physics of the model (same description as given in the following section for V4). The assimilation system is updated with new altimetry data from the instrument SWIM-nadir of CFOSAT mission.

Description of the changes in the upgraded V4 :

The CMEMS-GLO-WAV is upgraded with major improvements. Firstly, the model computing grid resolution decreases to 10 km and the minimum water depth decreases from 10 m to 5 m. Secondly, to improve the surface stress at the air/sea interface for high and intermediate winds, the dissipation source terms in model MFWAM (Meteo France WAve Model) are adjusted. Also, a diagnostic Philipps spectrum is now used for the high frequency part of the wave spectrum. The upgraded V4 wave system is forced by surface currents provided by the global PSY4-CMEMS ocean forecasting system with a daily update. Furthermore, V4 uses assimilation of directional wave spectra based on level 2 SAR (Synthetic Aperture Radar) wave spectra from Sentinel-1A and 1B. The assimilation of altimeter wave data is upgraded with the use of level 3 (L3) significant wave heights from Sentinel-3A.

Description of the previous system (V3):

The global wave system of Météo-France is based on the wave model MFWAM which is a third-generation wave model. MFWAM uses the computing code ECWAM-IFS-38R2 with a dissipation term developed by Ardhuin et al. (2010). MFWAM was upgraded on November 2014 with improvements obtained from the European research project “My Wave” (Janssen et al. 2014). The model mean bathymetry is generated from 2-minute gridded global topography data ETOPO2/NOAA. The native model grid is irregular with decreasing zonal distances close to the poles. At the equator, the zonal distance between grid points is fixed at 1/5°. The operational model, MFWAM, is driven by 6-hourly analysis and 3-hourly forecast winds from the IFS-ECMWF (European Center for Medium-Range Weather Forecasts) atmospheric system. The wave spectrum is discretized into 24 directions and 30 frequencies

from 0.035 Hz to 0.58 Hz. MFWAM assimilates altimetry data every 6 hours from Jason 2 & 3, Saral, and Cryosat-2. The global wave system provides analysis 4 times a day, and a 5-day forecast daily at 00:00h UTC. MFWAM uses partitioning to split the swell spectrum into primary and secondary swells. The output consists of 3-hourly wave products. The validation is conducted for significant wave height and peak period.

III VALIDATION FRAMEWORK

The quality of the global wave system MFWAM V4.12 (update of the assimilation system with SWIM-nadir data) has been assessed by comparison with wave observations from altimeter missions (HY-2A and HY-2B) and buoys during several months in 2020 (Table 2).

The quality of the global wave model has been assessed by comparison with wave observations provided by satellite altimeters (HY-2A and S3A) and buoys. The validation has been implemented for analysed significant wave heights and peak wave period. The altimeters HY-2A and 2B are not used in the assimilation, and it is then used for the validation of the analysis of significant wave height. The buoys wave data have been downloaded from the database of the JCOMM (Joint Commission for Oceanography and Marine Meteorology) monthly intercomparison developed at ECMWF.

The verification of the global wave system is thus twofold:

- The first part consists in comparing the significant wave height of the model with altimeters data. The colocation between the model and the altimeters is implemented with a time window of 3 hours. This gives globally a good coverage of all ocean basins.

- The second part consists in comparing the significant wave height and the peak wave period of the wave model with available buoys data provided by the database of the monthly JCOMM intercomparison developed at ECMWF (see Figure 1 for the buoys locations). More buoy data is used in the validation including data from Australian and Asian buoys. However, not all buoys measure peak period.

Table 2: Table describing the metrics and tests done to evaluate the quality of the product.

Name of metric	Variable	Metrics	Units	Reference dataset
SWH-Y-CLASS2-MOOR-BIAS/RMSD SWH-Y-CLASS1ALT-BIAS/RMSD	Significant wave height	2D maps, global average and regional average of mean bias and RMSD (Root Mean Square Deviation)	m	Satellite altimetry from CMEMS And In situ observations from buoys from JCOMM
SWH-Y-CLASS2-MOOR-SI SWH-Y-CLASS1-ALT-SI	Significant wave height	2D maps, global average and regional average of normalized scatter index = RMSD/MEAN	%	
WTPK-Y-CLASS2-MOOR-BIAS/RMSD WTPK-Y-CLASS4-ALT-BIAS/RMSD	Peak wave period	2D maps, global average and regional average of Bias and RMSD	seconds	In situ observations from buoys from JCOMM
WTPK-Y-CLASS2-MOOR-SI WTPK-Y-CLASS4-ALT-SI	Peak wave period	2D maps, global average and regional average of normalized scatter index = RMSD/MEAN	%	

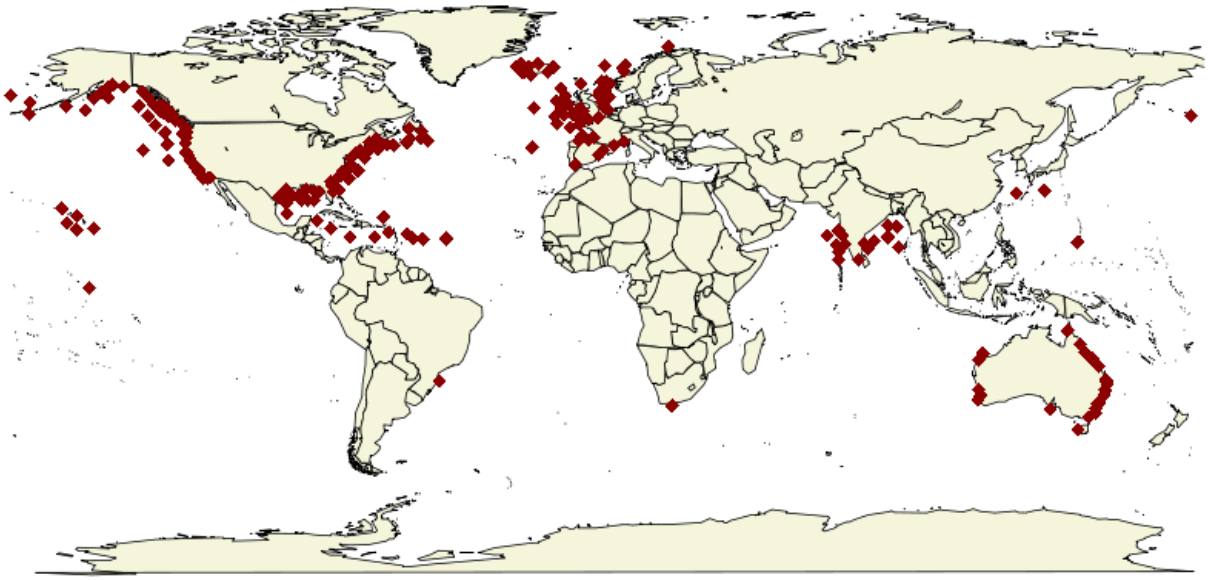

Figure 1: Locations of buoys downloaded from the monthly JCOMM intercomparison data base in 2020.

IV VALIDATION RESULTS

IV.1 Results related to update with SWIM wave spectra and S6A SWH (V4.2)

SWH from independent altimeters HY2B and 2C

The assimilation system CMEMS-GLO-WAV was updated with directional wave spectra from SWIM instrument of CFOSAT missions and SWH from Sentinel-6A altimeter mission. The tandem phase of Sentinel-6A and Jason-3 has shown the good quality of SWH from Sentinel-6A and inducing the impact of the assimilation of the same level as Jason-3. The validation of the update in CMEMS-GLO-WAV has been implemented with independent wave data from HY2B and HY2C missions. This latter has been calibrated particularly for HY2C which has tendency to overestimate high waves. The increasing number of assimilated data shows an improvement of scatter index of SWH which is in average of 8%, and small bias of 3 cm. We have also noticed a significant reduction of SWH bias in Southern Ocean, because of using SWIM directional wave spectra to better scale the wind-wave growth in unlimited fetch storm conditions. Only the results of the comparison against HY2A and 2B are presented in this document.

IV.2 Results related to update with SWIM-nadir (V4.1)

SWH from independent altimeters HY2A and 2B

The assimilation system CMEMS-GLO-WAV was updated with SWH provided by the SWIM instrument at nadir beam of CFOSAT satellite. The calibration/validation phase of CFOSAT has shown the good accuracy of SWH retrieved from nadir beam. In order to evaluate the impact of this update on the global wave system, the assimilation is compared with and without the update of SWIM with independent altimeter HY2A and HY2B. Three cycles of 13 days have been selected in February, June and August 2020 for this validation. The results indicate the good improvement of scatter index of SWH after updating with SWIM, on average by 7%. Figure 2 shows the scatter index of SWH for different ocean regions. The best reduction of SI (Scatter Index) is obtained at high latitudes where the uncertainties of wind forcing are stronger. Furthermore, the scatter index of SWH is below 8% in the tropics where swell waves tend to dominate.

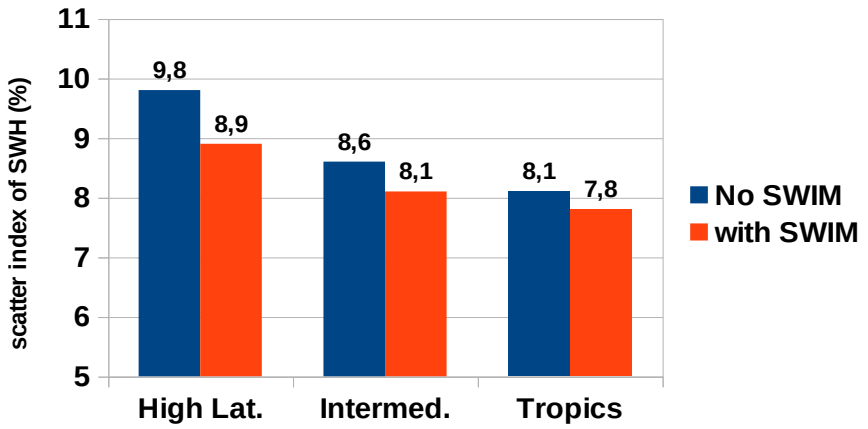

Figure 2: Scatter index of SWH from the CMEMS-GLO-WAV system in comparison with independent altimeter data from HY2A and HY2B in high, intermediate, and tropical latitudes. Blue and red colours stand for without and with SWIM-nadir data in the assimilation respectively.

To evaluate the impact of updating the assimilation system of CMEMS-GLO-WAV with SWH provided by SWIM-nadir, a validation has been performed for the winter season of 2020. SWH from independent HY2A & 2B altimeters have been co-located with CMEMS-GLO-WAV output. Figure 3 and Figure 4 show the bias and SI global distribution that indicate the good accuracy for most of the ocean regions. SI of SWH is smaller than 8% in open oceans and increases in the coastal areas. The bias is lightly positive and remains with values smaller than 10 cm. The bias increases in the Southern Ocean area affected by sea ice.

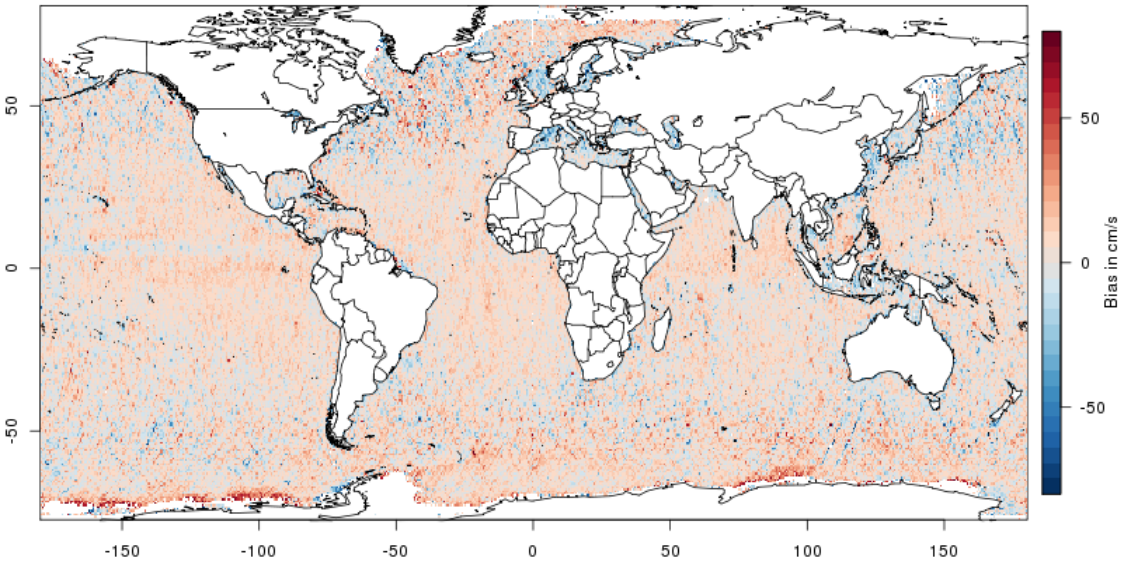

Figure 3: Global bias map of SWH from CMEMS-GLO-WAV with SWIM-nadir data for Jan-Feb-Mar 2020. The comparison is performed with HY2A and HY2B altimeter data.

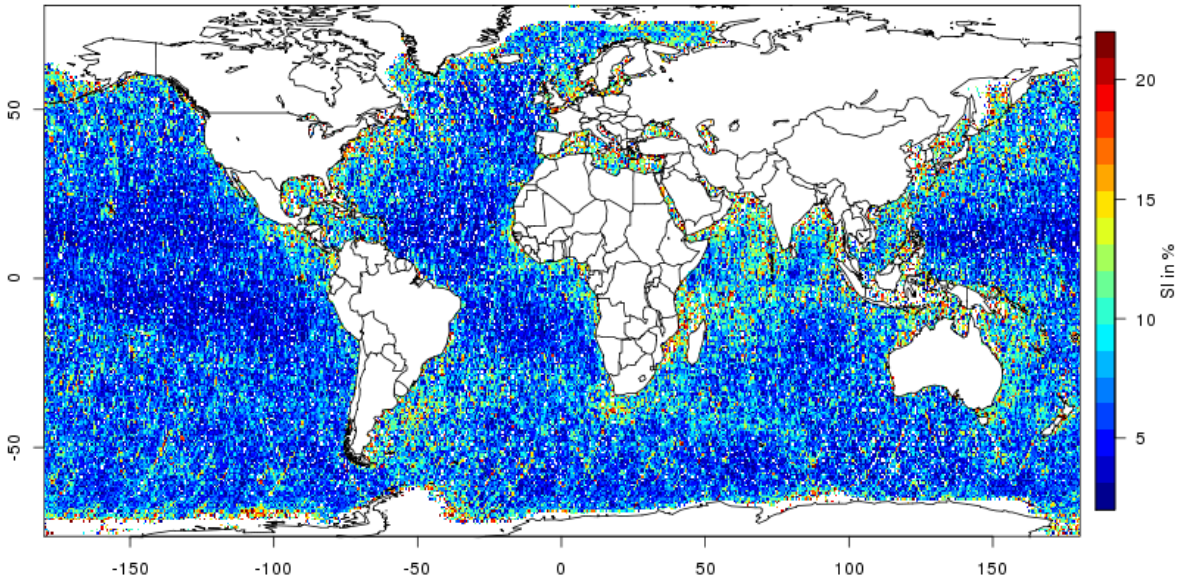

Figure 4: Global map of scatter index of SWH from CMEMS-GLO-WAV with SWIM-nadir data for Jan-Feb-Mar 2020. The comparison is performed with HY2A and HY2B altimeter data.

SWH and Peak period from NDBC buoys

The assimilation of SWIM data in CMEMS-GLO-WAVES has been evaluated during the period from January to May 2020, with NDBC (National Data Buoy Center) buoys mostly located in North America (Figure 1). The results indicate good accuracy in terms of scatter index with 13.8% for SWH and 12.6% for Tp (Table 3), while bias for SWH and peak period are 2 cm and 0.33 s, respectively.

Table 3: Comparison with NDBC buoys during the period January to May 2020.

	SWH	**Tp**
Bias	0.02 m	0.33 s
Scatter index (%)	13.8	12.6

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section summarizes the major change of the global wave system.

Table 4: Major change of the global wave system

Date	Change/Event description	System version	other
01/2018	Upgraded version of MFWAM model	V4	Adjustment of the model physics for coupling processes and upgrade of the assimilation system
12/2020	Update computing code and use of SWIM-nadir data in the assimilation system. The extension of wave forecast to 10 days.	V4.1	Use GRIB_API libraries and update of computing code
09/2022	Update of the assimilation system with S6A SWH and SWIM wave spectra of CFOSAT	V4.2	Adjustment of the assimilation

VI QUALITY CHANGES SINCE PREVIOUS VERSIONS

The global scale estimated accuracy numbers are based on comparisons with in-situ wave observations from buoys, from March to December 2016 for V4.1 and January to May 2020 for V4.2.

Table 5: EANs for GLOBAL_ANALYSIS_FORECAST_WAV_001_027 for last versions.

Variable	Metrics	Units	V4.1	V4.2
Significant wave height	Global average of bias	cm	1.0	2.0
	Global average of normalized scatter index	%	14.1	13.8
Peak wave period	Global average of bias	seconds	0.07	0.33
	Global average of normalized scatter index	%	10.7	12.6

Quality changes related to the update with SWIM wave spectra and Sentinel-6A SWH (V4.2)

This section gives a brief assessment related to the performance of CMEMS-GLO-WAV after the update of the assimilation of SWIM directional wave spectra and SWH from Sentinel-6A. The comparison with SWH from independent altimeters missions HY2B and HY2C has shown a significant improvement in all ocean basins. The bias of SWH in high latitudes, mid latitudes, and the tropics is of roughly 2,2 and 5 cm respectively. Figure 5 indicates the SWH bias map over all ocean regions. Overall, the bias is small and larger in coastal areas and in region, affected by sea ice. Figure 6 indicates the scatter index of SWH in high latitudes, mid latitudes, and the tropics of roughly less than 8% in average. This achievement highlights the reliability of the CMEMS-GLO-WAV system for ensuring the best wave products for users in their applications. Long term assessment with in-situ observations will be implemented in the coming month.

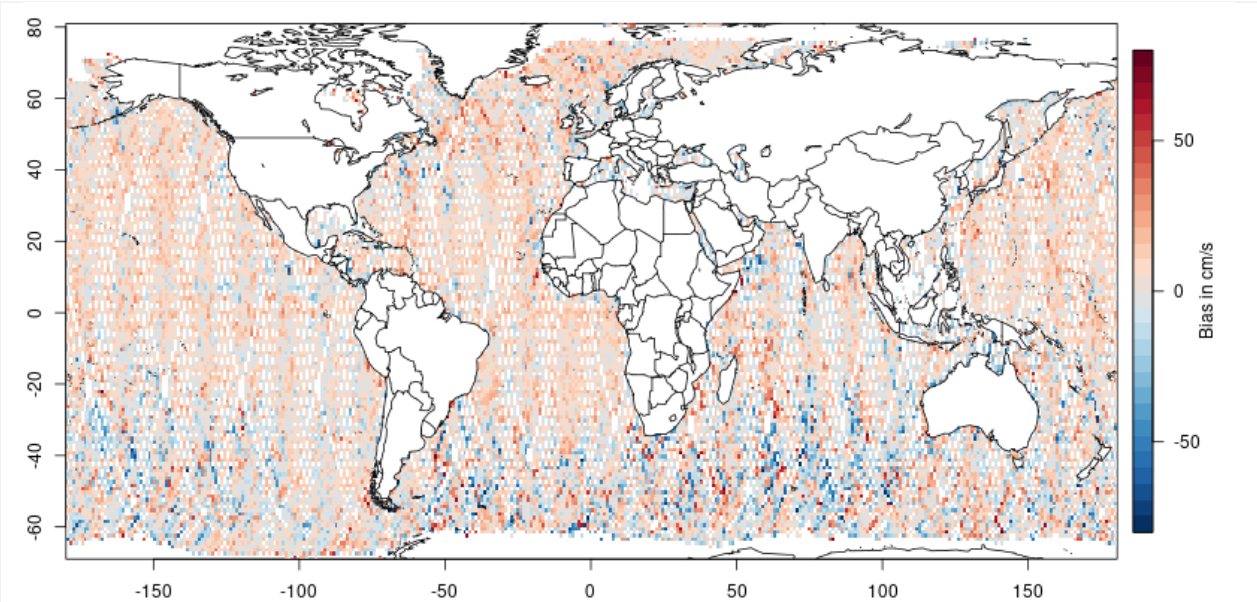

Figure 5: Bias map of SWH from CMEMS-GLO-WAV in comparison with HY2B and HY2C during the period starting from 14 to 30 June 2022. The maximum bias range is 60 cm. HY2B and HY2C are calibrated with reference mission Jason-3.

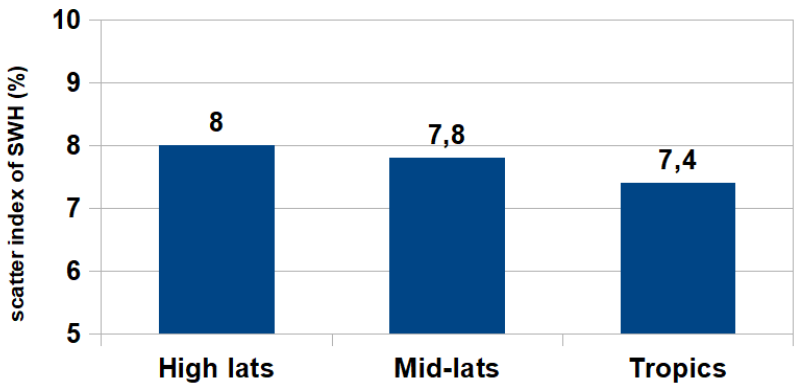

Figure 6: histogram of scatter index of SWH from CMEMS-GLO-WAV in high latitudes, mid latitudes and the tropics ocean regions. The comparison is performed with HY2B and HY2C independent altimeters missions from 14 to 3à June 2022.

Quality changes related to the use of SWIM-nadir data (V4.1)

This section gives a brief assessment related to the use of SWIM-nadir data in the assimilation system of CMEMS-GLO-WAV. Results are mostly provided by the validation with independent altimeter data from HY2A and HY2B missions. The scatter index of SWH from CMEMS-GLO-WAV is globally of 9%. The bias is small with an average of 4 cm. The use of SWIM-nadir data has significantly improved the scatter index of SWH during the 1-day forecast. The SI is reduced from 11.3% for February 2019, to 10.2% for February 2020.

Quality changes related to V4 version

This section aims to indicate the improvements induced by the CMEMS-GLO-WAV V4 system regarding the previous V3 system. The comparison between V4 and V3 has been performed with wave observations provided by the altimeter HY-2A and buoys (off-shore and coastal) for March, April and May 2017. The validation has been implemented for significant wave heights and peak wave period. Specific analysis has been developed by considering coastal buoys among the JCOMM archive (Figure 1 shows the locations of open oceans and coastal buoys).

Significant wave height:

The model analysis are compared to significant wave height (SWH) from the altimeter HY-2A (not included in the assimilation). The impact of the V4 system induces an improvement of the scatter index for all ocean basins, as illustrated in Figure 7. The largest reduction of the scatter index is obtained for the tropical ocean region. The results showed also the slight reduction of the bias of SWH for the intermediate latitudes from -15 to -13 cm, and the tropics from -9 to -6 cm.

The validation of the V4 system with SWH of JCOMM buoys shows an improvement of the scatter index by roughly 10 % in comparison with V3, as indicated in Figure 8. However, the analysed SWH from the V4 system increases the bias which remains small between 2 and 5 cm, as indicated in Figure 9. Table 1 shows the statistical analysis with the coastal buoys. This shows that the V4 system induces a significant improvement of the scatter index by roughly 19 % regarding to the V3 system; thanks to the better

resolution in the upgraded system. The bias from the V4 system is also reduced to zero and the correlation is well improved regarding to V3 system.

Peak wave period:

The V4 system shows a significant improvement of the scatter index of the peak period in comparison with the V3 system during the spring of 2017 as illustrated in Figure 10. Further, Figure 11 shows the monthly bias of the peak period, which reveals the significant reduction of the bias induced by the V4 system. For instance, in April 2017 the bias decreased from 0.29 to 0.06 seconds.

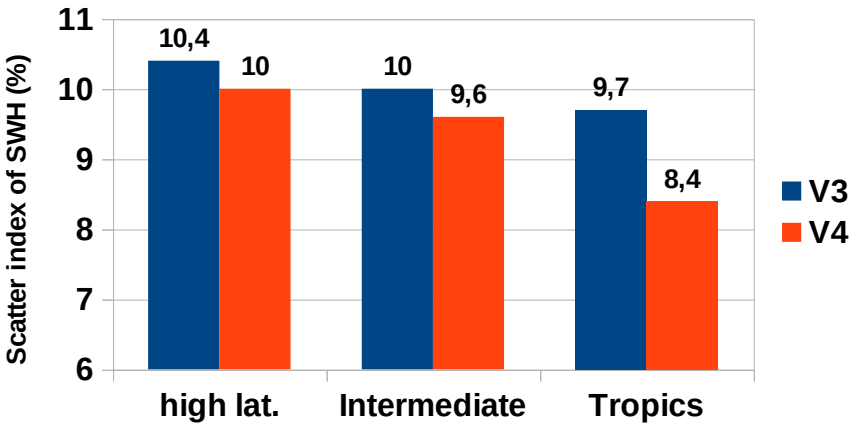

Figure 7: Histograms of normalized scatter index of SWH for high latitudes (|latitude >50°), intermediate latitudes (20°< latitude <50°) and tropics (latitude <20°). The comparison is performed with SWH from the Chinese altimeter HY-2A during the months of March, April, and May 2017. Red and blue colours indicate V4 and V3 systems respectively.

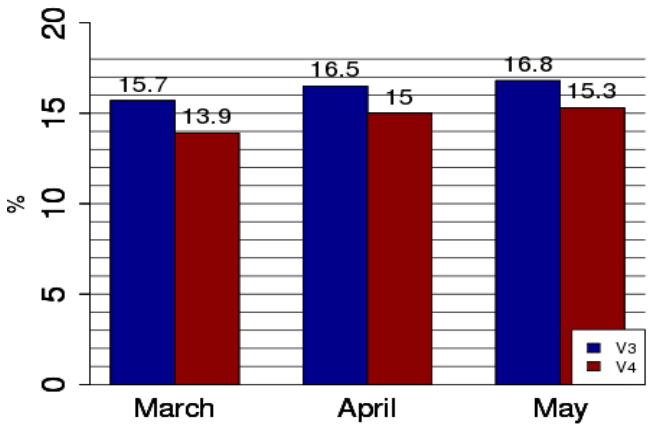

Figure 8: Monthly scatter index of SWH from buoys for March, April and May 2017. Red and blue colours indicate V4 and V3 CMEMS-GLO-WAV systems respectively.

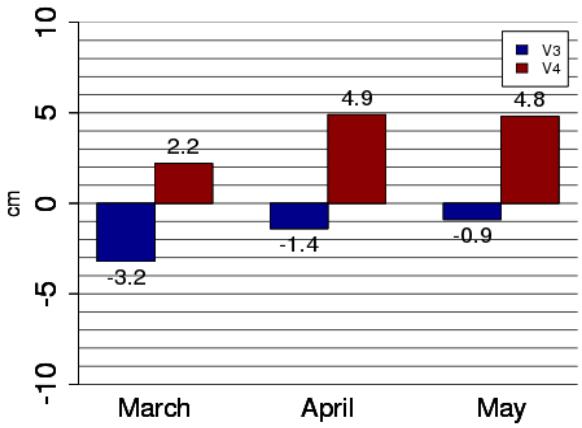

Figure 9: Monthly bias (in cm) of SWH from buoys for March, April and May 2017. Red and blue colours indicate the V4 and V3 CMEMS-GLO-WAV system, respectively.

Table 6: Statistical parameters of SWH from the comparison with the French coastal buoys (see locations in Figure 1) for March, April and May 2017.

Nb of data = 10177	**V3**	**V4**
Bias (m)	-0,03	0
RMSD (%)	24,1	19,4
Scatter index (%)	24	19,4
Correlation (%)	86,5	91,2

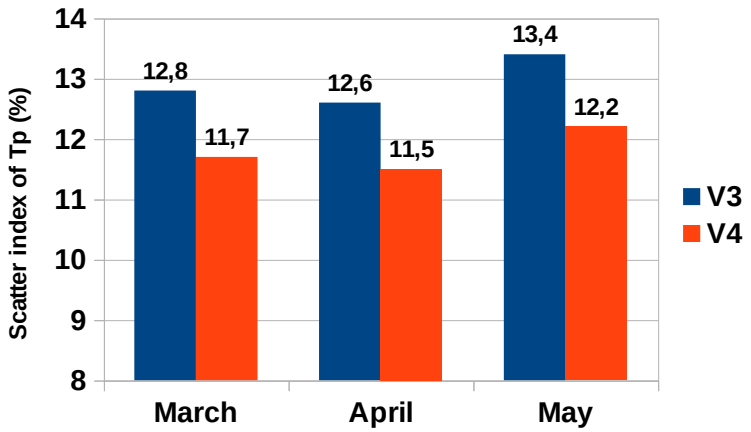

Figure 10: histogram of scatter index of peak period for March, April and May 2017. Red and blue colours indicate the V4 and V3 CMEMS-GLO-WAV systems respectively.

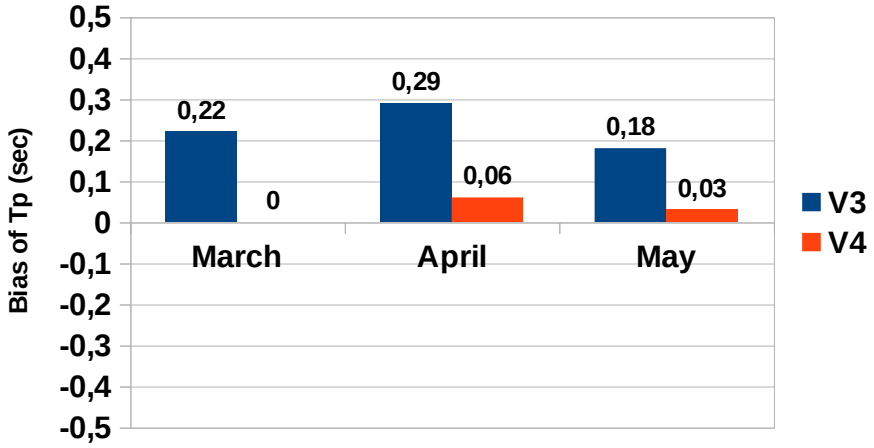

Figure 11: histogram of bias of peak period for March, April and May 2017. Red and blue colours indicate the V4 and V3 CMEMS-GLO-WAV systems respectively.

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